

CIS 4270/5270: Trustworthy Machine Learning

Spring 2026: Course Project Final Report

Dorm Chef: Teaching GPT to Cook with Constraints

Team: Akshat Bokdia, Rashi Agrawal, Ritika Agarwal

Introduction

We propose a constraint-following recipe generation task. Given a list of ingredients, the model must produce a quick (10-15 minute), student-friendly recipe that uses the listed ingredients, follows the desired cuisine or restriction, has 3-7 clear steps, and is realistic for a dorm kitchen. The task is well-suited to RL because correctness depends on multiple, partially independent constraints that the model must learn to satisfy jointly.

Training data

We generated **2,200** synthetic samples using GPT-4.1-mini and split it into **1,600 training**, **400 validation** and **200 test** sets. Each sample is an (ingredient list, preferred recipe, rejected recipe) triple. Ingredient lists are produced by seeding the generator with a (cuisine, meal type, dietary hint) tuple to ensure diversity. The preferred recipe satisfies all task constraints and the rejected recipe is generated under a separate system prompt that intentionally violates one constraint (extra ingredients, excessive cook time, too many steps, or unrealistic equipment). An LLM-as-judge sanity check filters samples whose preferred recipe does not satisfy the constraints. The same shared pool is reformatted into method-specific files: SFT uses (x, y); DPO uses (x, y, y').

Evaluation Metrics

We evaluate every model with a single LLM-as-judge harness (judge: GPT-4.1-mini, temperature 0). Hard metrics measure explicit constraint satisfaction; soft metrics measure overall quality.

- **Hard:** ingredient compliance, step count in range (3-7), cook time in range (≤ 15 min), constraint-followed (cuisine / dietary / exclusion / taste constraints honored), and the all-pass rate (all four hard constraints simultaneously).
- **Soft:** coherence, student-friendly, and an average LLM quality score combining both.
- **Descriptive:** average number of steps and average cook time across generations.

Base Model

Our base model is GPT-4.1-nano, evaluated in a zero-shot setting at temperature 1.0 (max_tokens 1024) with the system prompt: *"You are a recipe assistant for college students. Generate quick recipes (under 15 minutes) with 3-7 clear numbered steps, using only the ingredients provided."* On the original 400-sample validation set, the base model reaches 98.25% all-pass and an average LLM quality score of 1.0, leaving little headroom for fine-tuning to improve and motivating the harder stress-test evaluation that is the focus of this report.

Supervised Fine-Tuning (SFT)

SFT training uses the (x, y) view of the shared pool: 1,600 (ingredient list, preferred recipe) pairs for training and 400 for validation. We ran a six-configuration sweep over epochs, batch size, and learning-rate multiplier on GPT-4.1-nano. Run 3 (ep=3, bs=8, lr=0.5) was selected as SFT-alpha based

on its lowest training loss (0.217) among the six runs, with the validation curve held steady throughout training.

Direct Preference Optimization (DPO)

DPO training uses the (x, y, y') view of the same pool. Our six-configuration sweep added β as a third hyperparameter alongside epochs and learning rate. Runs 5 and 6 reached suspiciously low loss (0.04-0.09), a clear overfitting signal in DPO where high β or learning rate memorizes the preference pairs. We selected Run 6 (ep=2, bs=8, lr=0.5, β =0.1) as DPO-alpha.

Easy-Test Set Results

On the original 400-sample validation set, the three models are essentially indistinguishable on the headline metrics. Based on the feedback received on the draft report, we summarize these results briefly here and devote the remainder of the report to the harder stress-test evaluation.

Model	Ingredient Compliance	All-Pass Rate	Avg LLM Quality Score
Base GPT-4.1-nano	99.25%	98.25%	1.00
SFT-alpha (Run 3)	98.25%	98.25%	1.00
DPO-alpha (Run 6)	97.50%	97.00%	0.995

All three models achieve 97-98% all-pass rate. SFT-alpha matches the base model on all-pass; DPO-alpha is within 1.25pp. Soft-quality metrics are saturated at the ceiling. The full per-method analysis and the complete six-run hyperparameter sweep tables are documented in our draft report. The takeaway from this saturation is that simple prompt engineering already captures most of the achievable performance. This is what motivated the harder stress-test design described next.

Experiment 2: Stress-Test Set Design

To expose differences between methods, we constructed a 200-example stress-test set by combining the original constraints with additional difficult constraints: enforced cuisine (e.g., a Mexican dish from pasta + black beans + cheddar), unusual ingredient combinations (peanut butter + pasta + yogurt + broccoli), adversarial dietary restrictions (“make it vegan” when chicken is in the ingredient list), exclusion targets (eggs are listed but the recipe must avoid them), low-calorie preferences, and explicit taste profiles (spicy / savory / tangy). A new evaluation metric, constraint-followed, was added to capture whether the model honored the cuisine / restriction / exclusion / taste constraint specifically. The remaining hard metrics (ingredient compliance, step count, cook time, all-pass) are computed identically to the easy-set evaluation.

Stress-Test Set Results on Existing Models

We first re-evaluated the three previously trained models on the stress-test set without any retraining. All three models drop substantially, but the size of the drop differs sharply across methods (Figure 1).

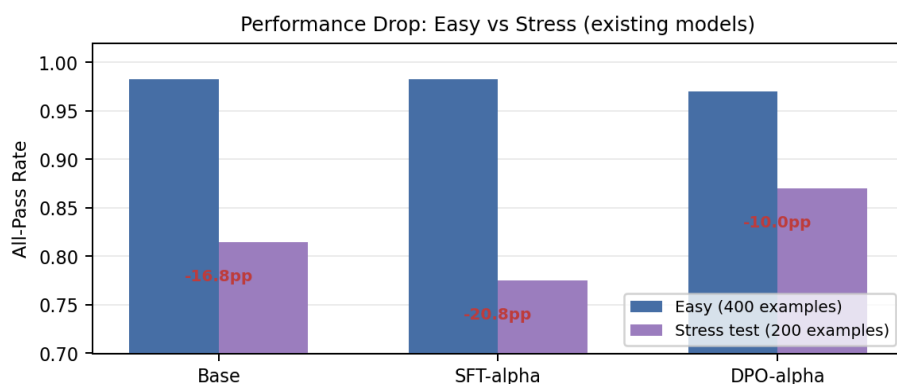


Figure 1. All-pass rate on the easy validation set (400 samples) versus the stress-test set (200 samples) for the three existing models. SFT-alpha shows the largest drop (~20.8 pp); DPO-alpha’s drop is the smallest (~10 pp). All three models are sensitive to the harder constraints.

Each method shows a different failure mode. SFT-alpha collapses on ingredient compliance, dropping from 98.25% on the easy set to 83.0% on the stress test-set (15-point drop) that takes it below the baseline model. However, SFT-alpha preserves perfect coherence (100%) and student-friendliness (100%), so the recipes it does produce remain well-formatted, plausible, and dorm-realistic. DPO-alpha, in contrast, retains 95.5% ingredient compliance and 87.0% all-pass under the harder constraints, but its coherence (98.5%) and student-friendliness (99.5%) are slightly below SFT-alpha’s. The base model sits between the two on most metrics.

Model	Ingredient Compliance	Cook Time	Constraint Followed	All-Pass Rate	Coherence	Student-Friendly
Base GPT-4.1-nano	95.00%	96.00%	87.00%	81.50%	98.00%	100.00%
SFT-alpha (easy-trained)	83.00%	100.00%	89.50%	77.50%	100.00%	100.00%
DPO-alpha (easy-trained)	95.50%	98.50%	90.50%	87.00%	98.50%	99.50%
SFT-hard (re-trained)	88.50%	99.50%	89.50%	81.00%	100.00%	100.00%
DPO-hard (re-trained)	93.50%	95.50%	93.50%	83.50%	95.50%	97.50%

These numbers support the framing that the two methods are learning different things. SFT learns to imitate the surface form of well-written recipes (clear formatting, plausible dishes, dorm-friendly language) which is why coherence and student-friendliness stay at 100% even when the model fails to use the right ingredients. DPO, having seen explicit preference contrasts during training, learns to reason about whether a candidate output satisfies the explicit constraints, which is why its constraint compliance is more robust to harder inputs. Neither behavior is strictly better; the right method depends on whether the priority is constraint enforcement or natural, coherent recipe writing.

Experiment 2.1: Retraining on Harder Data

We then trained two new models, SFT-hard and DPO-hard, on a synthesized 1,600-sample harder training set drawn from the same constraint distribution as the stress test (cuisine, taste, restriction, exclusion). To allow a controlled comparison, we reused the best hyperparameters from the easy-set sweep rather than running a new sweep.

Method	Hyperparameter Configuration	Final Train Loss	Final Full-Valid Loss	Num Steps
SFT-hard	ep=3, bs=8, lr=0.5	0.467	0.417	600
DPO-hard	ep=2, bs=8, lr=0.5, $\beta=0.1$	0.004	0.020	400

Note that DPO-hard’s loss (0.004 train, 0.020 valid) is much lower than what we observed on the easy data, similar to the overfitting region we saw at high β / LR in the easy sweep. We considered this a warning signal but kept the run for the comparison reported below.

Stress-test results across all five models

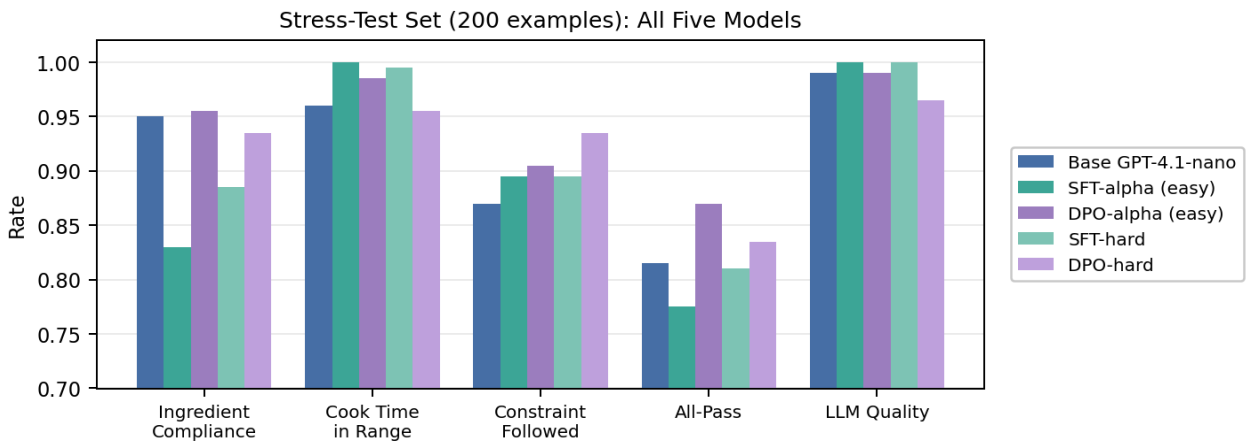


Figure 2. All five models on the 200-sample stress-test set. SFT models keep coherence and LLM quality at the ceiling; DPO models are stronger on constraint-following but lose ground on cook time, step count, and — for DPO-hard — LLM quality.

Retraining on the harder data exposes the trade-off cleanly. SFT-hard improves on ingredient compliance (83.0% → 88.5%) and gains 3.5 points of all-pass over SFT-alpha (77.5% → 81.0%). It also keeps coherence and student-friendliness at the ceiling (100% / 100%), preserving recipe quality. DPO-hard posts the highest constraint-followed score across all five models (93.5%, up from 90.5% for DPO-alpha); retraining on adversarial constraint data does sharpen DPO’s constraint-reasoning. However, it is also the only model whose coherence drops below 96.0% (95.5%) and the only model whose LLM quality score falls below 0.99 (0.965). Cook time in range and step count in range also drop relative to DPO-alpha (98.5% → 95.5%, 99.5% → 94.0%). DPO is increasingly trading recipe naturalness for constraint compliance.

This trade-off is the central finding of the report. There is no single best model: SFT produces consistently coherent, well-formatted, dorm-realistic recipes but is brittle to harder constraints; DPO honors the constraints more robustly but at the cost of fluency, especially after retraining on harder data.

Discussion

Our results support the framing that DPO and SFT learn different aspects of the task. DPO consistently improves constraint reasoning: the easy-trained DPO-alpha already outperforms SFT-alpha on the constraint-followed metric at the stress-test boundary (90.5% vs 89.5%), and DPO-hard pushes this further to 93.5%. SFT, in contrast, makes its largest absolute gains on surface compliance: SFT-hard improves ingredient compliance over SFT-alpha by 5.5 percentage points and keeps coherence and student-friendliness at the ceiling, even on the harder evaluation.

The trade-off is most visible at the extreme end of constraint training. DPO-hard achieves the strongest constraint-followed score in the study, but its coherence falls to 95.5% and its LLM quality score to 0.965 and it is the only model below 0.99 on either metric. Strong preference optimization for constraint satisfaction comes at a measurable cost to recipe naturalness. For a recipe-generation system

intended for actual student use, that fluency matters: a coherent recipe that misses a constraint is often more useful than a constraint-perfect recipe that reads awkwardly.

Regarding metric saturation, the easy validation set produced an effectively identical 97-98% all-pass for all three models, suggesting both fine-tuning approaches were ineffective. The stress-test set widened that range to 77.5–87.0%. The methods were not actually identical; the original metric simply could not see the difference.

Conclusion and Future Work

We compared SFT and DPO on a constraint-following recipe generation task. On a saturated easy benchmark, all three methods were indistinguishable, but a harder stress-test set revealed a clear trade-off: DPO improves constraint reasoning (the highest constraint-followed score), while SFT preserves recipe quality (coherence, student-friendliness, and clean formatting) even when the harder constraints are missed. Neither method is uniformly better; the right choice depends on whether the priority for the deployed system is strict constraint enforcement or natural, well-written recipes.

Future work

- Scale and diversify the SFT training distribution; SFT's clearest weakness is generalization, which more diverse data should address.
- Tune the β parameter systematically for the harder DPO regime; our reuse of easy-sweep hyperparameters could have under-regularized DPO-hard, which may explain the observed overfitting and the loss of structural compliance.
- Run a proper RFT / online RL pass with a multi-grader that combines ingredient compliance, brevity, and constraint-following into a single reward; this would directly optimize for the metrics we care about, rather than relying on imitation or preference signals as proxies.